Approval-Based Elections in the 1D Euclidean Domain

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This dissertation is submitted for the degree of
Master of Science

December 2020

TODO

0.1 My Questions

1. Change order of 4.1(VCR Profiles Statistics) and 4.2(VCR Detection Algo)
2. Split Ch.4 (Solution) -> 4 (detection), 5 (properties), 6 (mav), 7 (deletion to vr / cr)
3. Profile / Preference naming convention

0.2 10/12/2020

1. Refactor code - vcr ilp detection, profile generation
2. Generate and analyze profiles: 5x5, 4x6, 6x4, random NxM
3. Implement ilp MAV
4. Implement VR/CR MAV Liu and Guo (2016)
5. Compare ilp and poly MAV

0.3 17/12/2020

1.

Abstract

(WIP)

Praca polega na opracowaniu algorytmów dla szeregu problemów wyborczych w jednowymiarowej dziedzinie euklidesowej (w modelu Voter/Candidate Range). Oprócz algorytmów, dla problemów NP-trudnych w większych dziedzinach preferencji, w ramach projektu będzie należało także:

- poszukiwać algorytmów rozpoznających jednowymiarowe dziedziny Euklidesowe na podstawie danych aprobatowych
- oszacować jak wiele wyborów należy do rozważanej dziedziny
- poszukiwać algorytmów stwierdzających jak podobne są dane wybory do pochodzących z dziedziny jednowymiarowej

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Chapter 1

Introduction

Elections are ubiquitous element of our society in one form or another. For example choosing members of parliament, winner of a competition. In a day-to-day activities we need to decide on a meeting time or select recommended movies in a streaming service. Each of these choices may require different type of election and scoring rule, based on the underlying properties of the choice (i.e. voters should be represented proportionally by members of a government)

(WIP) Write here about types of elections, single / multi winner, etc. , different scoring rules.

In this work I am going to focus on the *Approval Elections*, that is where each voter provides a subset of candidates that he approves of and disapproves the rest. In general many interesting voting rules are known to be NP-hard, however become polynomial solvable under some restrictions on the structure of the preferences.

Elkind and Lackner (2015) introduced a new one-dimensional domain restriction (VR / CR) and showed relation with existing ones (SP, SC and others). Faliszewski (2020) generalized the model (VCR) and explored two-dimensional case.

The goal of this work is to explore properties of the one-dimensional VCR model and find polynomial algorithms for detection and scoring rules.

Chapter 2

Preliminaries

2.1 Basic Definitions

In this section, I introduce the definitions and notations used in this work.

Approval Elections Election is a pair $E = (C, V)$ where $C = \{c_1, ..., c_m\}$ is a set of candidates and $V = \{v_1, ..., v_n\}$ is a list of voters. Each voter $v \in V$ approves a subset of the candidates and disapproves the remaining candidates, I will be referring to it as a *approval set* and for each voter $v \in V$ denote it as $A(v)$

Committee Selection Rules

For the detailed overview of multiwinner voting rules I point to Faliszewski et al. (2017)

Minimax Approval Voting (MAV) (WIP) Selects a committee W that minimizes the maximum Hamming distance between W and voters' approval ballots.

For an election E and a resulting committee of size k I denote winning committee as $W = MAV(E, k)$

2.2 (WIP) Related Work

- An Analysis of Approval-Based Committee Rules for 2D-Euclidean Elections by Faliszewski (2020)
- Overview of various domain restriction in Structure in Dichotomous Preferences by Elkind and Lackner (2015)
- Minimax Approval Voting rule introduced by Brams et al. (2007)
- Polynomial algorithms for MAV in VR and CR domains by Liu and Guo (2016)
- Example of a new domain restriction (SPOC) and recognition algorithms by Peters and Lackner (2020)
- Example of a NP-hard domain recognition algorithms and theorems by Peters (2017)
- Alternative approach to C1P for domain recognition by Magiera and Faliszewski (2019)

Chapter 3

Approval-Based Elections in the *1D* Euclidean Domain

3.1 Euclidean Preference Restrictions

Voter Candidate Range Given an election $E = (C, V)$, we say that the voters have *1D-Euclidean* preferences For each agent there exists a point x_a and nonnegative real value r_a such that: $c \in A(v) \Leftrightarrow \sqrt{(x_c - x_v)^2} \leq r_c + r_v$

Intuitively, for $a \in C \cup V$ the point x_a describes a's ideal position in 1-dimensional space of opinions. For candidate $c \in C, r_c$ can be seen as c's charisma. For candidate $v \in V, r_v$ can be seen as v's willingness to compromise. Faliszewski (2020)

Candidate Range We say that P satisfies CR if candidates in C can be ordered so that each of the sets vi forms an interval of that ordering. CR is special case of VCR, where we require all the voters to have radii equal to zero

Voter Range We say that P satisfies VR if the voters in P can be reordered so that for every candidate c the voters that approve c form an interval of that ordering. VR is a special case of VCR, where we require all the candidates to have radii equal to zero

CR and *VR* were introduced in Elkind and Lackner (2015) under the names *CI* and *VI*.

The VCR model is strictly more powerful than the VR and CR ones. For example, election with candidate set $C = \{a, b, c, d\}$ and voters with approval sets $\{a, b\}$, $\{b, c\}$, $\{b, d\}$, and $\{a, b, c, d\}$ is VCR, but neither VR nor CR.

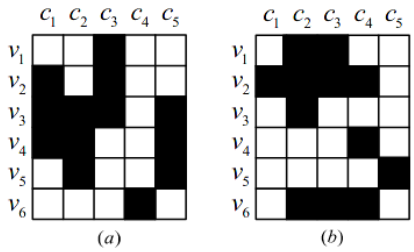

Fig. 3.1 VR and CR Temporary image

3.1.1 VR and CR Detection algorithms

Profile is VR or CR if the matrix or any permutation has consecutive ones property, that is for each row (VR) or column (CR), entries with 1 are in the uninterrupted sequence.

The fastest known algorithm for detection of C1P is based on the *PQ-tree* algorithms by Booth and Lueker (1976). However in this work I'm using still relatively fast and much easier to implement approach based on the *Lex-BFS* algorithm by Habib et al. (2000)

3.2 (WIP) Why VCR?

This is a problem statement. Explain how VR/CR/VCR can change complexity of computing rules, detection, etc.

Chapter 4

Solution

4.1 Voter Candidate Range Profiles

In this section I describe carried out experiments and their results. I've found various interesting features of the VCR profiles (WIP).

4.1.1 (WIP) Experiment description - search of domain

Experiments can be grouped into 2 basic categories: *exhaustive* and *random* search.
Steps were as follows:

1. Given size of the election C and V
2. Generate approval sets of the profiles
3. Filter out profiles not fullfilling *VCR* property. Described in the section 4.2
4. Group profiles by *VR, CR* and *VCR not VR nor CR* properties
5. Aggregate statistics

To achive high performance I used *MapReduce* paradigm with *Spark*, as the tasks are independent and easy to parallelize.

4.1.2 How many VCR profiles exists that are not VR nor CR?

Search of all the small profiles

For the smaller size of profiles I was able to perform search of all the profiles step 2 of 4.1.1.

	3C 3V	4C 4V	5C 5V	4C 6V	6C 4V
Count	512	65536	2^{25}	2^{24}	2^{24}
VCR	506	57832	-	-	-
VR	-	-	-	-	-
CR	-	-	-	-	-
VCR !COP	0	96	-	-	-

Search of the random large profiles

For the larger size of profiles only difference is in the step 2 of 4.1.1. Where I generated an suitable number of random profiles.

	20C 20V	60C 60V	100C 100V
Count	-	-	-
VCR	-	-	-
VR	-	-	-
CR	-	-	-
VCR !COP	-	-	-

4.1.3 (WIP) Experiment conclusions

Maybe do some nice graphs here

4.2 VCR Profile Detection Algorithm

This section describes in detail and compares created algorithms for the profile VCR property detection.

4.2.1 Integer Linear Programming

The only input is *ApprovalSet* of each voter.
Represented by $2D$ matrix A_{vc}

Variables

x_c - candidate c position on x axis
x_v - voter v position on x axis
r_c - candidate c radius
r_v - voter v radius

Transform VCR condition to Linear constraints

$$\begin{aligned} VCR = & \\ &= \sqrt{(x_c - x_v)^2} \leq r_c + r_v \\ &= \|x_c - x_v\| \leq r_c + r_v \\ &= x_c - x_v \leq r_c + r_v \; or - (x_c - x_v) \leq r_c + r_v \\ &= -(r_c + r_v) \leq x_c - x_v \leq r_c + r_v \end{aligned} \tag{4.1}$$

Positive and Negative Condition and Big M method

Introduce large constant positive parmater
M

Introduce new binary variables for each pair of candidate and voter:
$z1_{cv}$
$z2_{cv}$

$$
\begin{aligned}
&\textit{Minimize 0}\\
&\textit{foreach (c,v) in } C \times V:\\
&\textit{if } c \in ApprovalSet(v) \textit{ add constraints}\\
&x_c - x_v \leq r_c + r_v\\
&-(x_c - x_v) \leq r_c + r_v\\
&\textit{if } c \notin ApprovalSet(v) \textit{ add constraints}\\
&x_c - x_v + M(1 - z1_{cv}) \geq r_c + r_v + 1\\
&-(x_c - x_v) + M(1 - z2_{cv}) \geq r_c + r_v + 1\\
&z1_{cv} + z2_{cv} >= 1
\end{aligned}
$$

Mock Distance Variable Condition

Introduce new variable
D_{cv} - distance between candidate c and voter v on x axis

$$
\begin{aligned}
&\textit{Minimize 0}\\
&\textit{foreach (c,v) in } C \times V:\\
&\textit{add constraints}\\
&D_{cv} = x_c - x_v\\
&D_{cv} = -(x_c - x_v)\\
&\textit{if } c \in ApprovalSet(v) \textit{ add constraints}\\
&D_{cv} \leq r_c + r_v\\
&\textit{if } c \notin ApprovalSet(v) \textit{ add constraints}\\
&D_{cv} \geq r_c + r_v + 1
\end{aligned}
$$

4.2.2 Polynomial VCR detection algorithm

4.2.3 Performance Comparison

4.3 Minimax Scoring Rule Algorithms

4.3.1 Overview

Minimax Approval Voting (MAV) As close as possible to the preferences of all parties. By "close" we mean that the maximum distance of the compromise from the position of any state is minimal, which we call a minimax outcome. Brams et al. (2007) Selects a committee W that minimizes the maximum Hamming distance between W and voters' approval ballots. We define MAV-score of a committee W as $Max(f(W,v1),...,f(W,vn))$ where $f(W,vi) =$. MAV returns a committee W with the lowest MAV-score.

```
def hammingDistance(a: np.array, b: np.array) -> int:
return (a != b).nonzero()[0].shape[0]

def mavScore(profile: Profile, committee:np.array) -> int:
    partialHD = lambda preference: hammingDistance(committee, preference)
    return max([partialHD(pref) for pref in profile.A])

def minimax(profile: Profile, k: int) -> np.array:
    candidateCount = len(profile.C)
    scores = [(mavScore(profile, np.array(committee)), profile)
                for committee in combinations(range(candidateCount), k)]
    scores.sort(key=lambda mav_p: mav_p[0])
    return scores
```

Implementation of ILP from Lackner and Skowron (2020)
Implementation of Poly for VR and CR from Liu and Guo (2016)

4.3.2 Polynomial algorithm for VCR profiles

4.3.3 Comparison

4.4 TBD: Approval deletion for VCR profile to become VR or CR

Chapter 5

Evaluation

5.1 Section One

5.2 Section Two

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